

Technical Specification: Interactive Semantic Search Engine

1. Operational Overview (FR-4)

The Semantic Search Module (`semantic_search.py`) houses the system query logic, fulfilling Functional Requirement 4. The script receives arbitrary user text strings via an interactive runtime terminal interface, vectorizes the query data, and queries the PostgreSQL datastore using vector math.

2. Mathematical Approach: Cosine Distance

Traditional database queries rely on relational keyword indexes (B-Trees / GIN). This search interface utilizes the pgvector custom operator (`<=>`) to calculate the Cosine Distance between the query vector and document vectors. The true similarity score is calculated by subtracting the distance from 1:

Similarity Score = $1 - (\text{embedding} \text{<=> QueryVector})$

3. Execution Architecture

- **Vector Transformation:** Runs the user input query string through the initialized transformer network.
- **SQL Execution Block:** Connects to the database and runs a parameterized query using `ORDER BY` and `LIMIT 5` filters.
- **Formatting Engine:** Loops through the database return array, printing similarity margins, source document filenames, and matching text snippets.

4. Interface Lifecycle

Implements an infinite query cycle loop running inside the console environment. The interface processes user requests continuously and terminates gracefully when it catches an explicit 'exit' token command.